

Accelerating Enterprise Migrations to Databricks with Devin

How autonomous AI engineering compresses Snowflake → Databricks and Exadata → Databricks migrations for large regulated enterprises — banks, telcos and public-sector bodies — from months to days, accelerating time-to-platform and pulling Databricks consumption (DBUs, the platform's billing unit) forward.

98.8% COST REDUCTION VS. MANUAL LABOR	8 days 1 PLATFORM, DISCOVERY TO VALIDATED CUTOVER	50k+ ETL JOBS MIGRATED IN ONE PROGRAM	99% MIGRATION COMPLETENESS
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■ WHAT DEVIN IS

Devin is an autonomous AI software engineer from Cognition. Unlike a code assistant that suggests snippets, Devin plans and executes entire engineering tasks end-to-end: it ingests legacy artifacts, maps dependencies, writes idiomatic target code, runs it, diagnoses failures, and iterates until the output is validated. What makes Devin uniquely suited to platform migrations is **parallel scale** — each job runs in its own isolated session, and hundreds run at once, governed by reusable playbooks that improve with every job. Critically for a regulated enterprise, **Devin does the engineering while humans retain review, approval and sign-off**: every change lands as a reviewed pull request with a full audit trail, so migration velocity never bypasses the organisation's change controls.

■ WHERE DEVIN ACCELERATES THESE MIGRATIONS

Large regulated enterprises typically run two migration tracks toward Databricks — Snowflake and Oracle Exadata. Both are high-effort, high-repetition programs where the engineering labor, not the platform, is the bottleneck. Devin attacks the labor directly, and knows where the hard parts live in each:

Snowflake → Databricks • Types & semantics: VARIANT / JSON (FLATTEN→exp lode, dot-notation→nested structs), NUMBER precision/scale, TIMESTAMP_NTZ/LTZ/TZ & timezone handling, sequences→identity. • Programmable objects: SQL/JS/Python UDFs & UDTFs, stored procedures, Snowpark→PySpark, external functions. • Streams & Tasks → Structured Streaming / Delta Live Tables (DLT); MERGE-based CDC semantics preserved. • Views, materialized/secure views, time-travel, zero-copy clones → Delta equivalents (gaps flagged where no 1:1 mapping exists). • Governance: RBAC grants, row-access & dynamic-masking policies → Unity Catalog privileges, row filters, column masks. • Storage: stages / integrations → UC Volumes & external locations. • Profiling: ACCOUNT_USAGE / query history to prioritize by cost & consumers.	Exadata / Oracle → Databricks • PL/SQL: packages, cursors, exception blocks, triggers, sequences, autonomous transactions, CONNECT BY hierarchies, MODEL clause. • Oracle-isms: DECODE, NVL/NVL2, ROWNUM, analytic functions, RAW/CLOB/BLOB, DATE-vs-TIMESTAMP, NLS/charset. • Physical→logical: partitioning, indexes (incl. bitmap) & IOTs → Delta partitioning, liquid clustering, Z-order + OPTIMIZE. Delta has no secondary indexes, so the access strategy is redesigned, not copied. • Constraints: Delta does not enforce PK/FK/unique — integrity is instead proven via reconciliation (below). • CDC: GoldenGate / redo-log capture → Auto Loader / DLT, distinguishing initial bulk load from ongoing incremental.
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■ DEVIN AND DATABRICKS' OWN AGENTS — COMPLEMENTARY, NOT COMPETING

Databricks' native tooling already automates much of the mechanical conversion. Devin works one layer up — as an autonomous engineer that executes the entire project to closure — so the two compound rather than overlap. The heavy lifting Devin owns is precisely the human-engineer labour the tooling leaves behind.

What Databricks tooling already does

- **Lakebridge** (Analyzer / Converter / Validator): profiles the source estate, transpiles SQL & ETL dialects to Databricks, and provides reconciliation tooling — automating much of the mechanical conversion.
- **Agent Bricks, Genie, Databricks Assistant**: build GenAI apps, answer analytics questions, and assist in-notebook — aimed at applications and analytics on the platform, not at running a legacy migration.

Where Devin does the heavy lifting

- **Autonomous execution to closure**: runs converted code, diagnoses failures and iterates until it passes — the hard last ~20% a transpiler hands back to engineers.
- **Semantics, not just syntax**: reverse-engineers undocumented business logic and resolves constructs with no 1:1 mapping.
- **The whole estate**: migrates surrounding orchestration, CI/CD and consumer repointing — not just SQL conversion.
- **Parallel scale to sign-off**: hundreds of isolated sessions across thousands of jobs, each validated to a reconciliation report, with human approval gates.

Devin consumes Databricks assessment / conversion output and produces Databricks-native assets — it accelerates the very platform it is migrating onto.

■ DATA MOVEMENT & SECURE CONNECTIVITY

Devin extracts through partitioned JDBC reads, staging large tables as Parquet in cloud object storage before loading to Delta. Connectivity stays inside the customer's approved boundary — Private Link / VNet peering, credentials in a secrets manager, no data leaving the sanctioned network — and separates a one-time bulk historical load from ongoing incremental capture, so cutover does not require freezing the source.

■ VALIDATION & RECONCILIATION

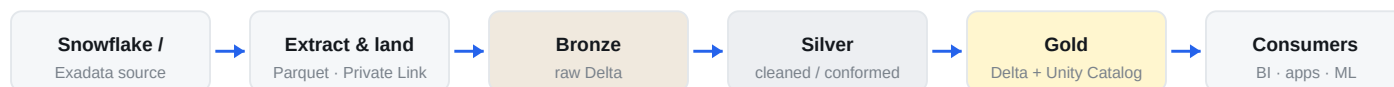
Correctness is proven, not spot-checked. Devin generates and runs an equivalence framework, then produces a reconciliation report as the sign-off artifact:

- Row counts **plus** column-level aggregate checksums (SUM/MIN/MAX and per-column hashes), null and distinct-value profiling.
- **Financial exact-match** reconciliation on monetary tables; documented decimal/float tolerance for non-exact columns.
- Ordering / non-determinism and timezone normalization handled, plus a deterministic sampling strategy for very large tables — coverage stated in the report.

■ ORCHESTRATION, ARCHITECTURE & GOVERNANCE

- **Medallion architecture** (bronze = raw, silver = cleaned / conformed, gold = business-ready): Devin builds all three layers — the plan below makes the silver layer explicit rather than jumping bronze→gold.
- **Scheduler conversion**: Airflow, Control-M and Oracle Scheduler dependency graphs → Databricks Workflows / DLT pipelines, preserving the DAG.
- **Unity Catalog design**: catalog / schema layout, lineage capture, and a least-privilege access model.
- **CI/CD**: Databricks Asset Bundles / Terraform, Git, and dev→test→prod promotion — production-grade, not one-off notebooks.
- **Performance & cost**: Photon, serverless, cluster policies, file compaction / OPTIMIZE, and Z-order / liquid clustering.

■ TARGET ARCHITECTURE



DEVIN builds, tests and reconciles every stage — humans review & approve each pull request

■ GOVERNANCE & COMPLIANCE

The approach is built for regulated environments: data stays in-region for **residency / sovereignty**; PII is masked and encrypted; all access and transformations are audit-logged; and the workflow aligns with sector regulation (e.g., **APRA CPS 234**, GDPR, local data-sovereignty rules) and the organisation's change-management, with **human-in-the-loop approval gates** at each

promotion. Devin accelerates the engineering; the customer retains control of what ships.

■ PROVEN AT ENTERPRISE SCALE

Devin has already run migrations of exactly this shape — including in banking — with measured, audited results. In an anonymized insurer's warehouse migration to Databricks Unity Catalog, Devin autonomously executed schema creation, ETL development, data-quality resolution and validation, deploying 15 notebooks and 5 scheduled jobs across 142 database objects. Comparable programs (anonymized) include a **Fortune-10 healthcare** estate (50,000+ ETL jobs and 10,000+ Airflow DAGs, years compressed to weeks), a **top-5 global bank** (1,300 ETL jobs at ~20× cost savings), a **major Latin-American bank** (platform migrations 6× faster), and a **global automotive OEM** (a 6-month migration delivered in 3 weeks).

DIMENSION	TRADITIONAL APPROACH	WITH DEVIN
Timeline	2–3 months (full team)	8 days (1 resource, part-time)
Cost	\$165,000 estimated	~\$2,000 actual
Data accuracy	Manual spot-checks	99%+ validated; financial tables exact match
Issues resolved	Sequential, engineer-driven	20+ autonomously diagnosed & fixed

■ HOW AN ENGAGEMENT RUNS

PHASE	ACTIVITY
Days 1–2	Discovery & scoping — profile query logs, map consumers, catalog objects, agree the reconciliation bar.
Days 3–4	Bronze & silver — extract and land data, convert DDL to Delta, deploy CDC / Auto Loader, conform to silver.
Days 5–6	Gold & orchestration — build transformation logic, convert schedulers to Workflows / DLT, register in Unity Catalog.
Days 7–8	Validation & hardening — run reconciliation across all objects, resolve issues, produce sign-off report.
Cutover	Parallel dual-run + reconciliation window, repoint BI / consumers, execute cutover with a documented rollback, then decommission source.
Week 2+	At scale — parallel conversion of thousands of jobs via configurable playbooks.

■ ASSUMPTIONS & RISK

ASSUMPTIONS — WHAT THE CUSTOMER / DATABRICKS PROVIDE	RISKS & MITIGATIONS
• Read-only source access & credentials (Snowflake / Exadata). • Network connectivity into a non-prod Databricks workspace. • A representative sample of workloads to scope the pilot. • A named customer counterpart for logic questions & sign-off.	• Undocumented business logic → Devin reverse-engineers & documents it for stakeholder confirmation. • Proprietary source features with no 1:1 mapping → flagged early with a proposed equivalent. • Source data-quality issues → surfaced by profiling / reconciliation rather than silently carried over.

■ RECOMMENDED NEXT STEP

Scope one pilot migration — a representative Snowflake or Exadata workload — to demonstrate a validated cutover and set the reusable playbook for scale.
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What takes a team months,
Devin compresses to days.